

Lab - 2: Investigating Network Traffic over HTTP vs. HTTPS

Objective

To analyze and compare the transmission of sensitive information (login credentials) over unencrypted (HTTP) and encrypted (HTTPS) protocols using Wireshark.

Phase 1: HTTP Traffic Analysis (Unencrypted)

1. The Login Process

The investigation begins with a user accessing the "VulnLab Bank Login" portal over a standard HTTP connection. As seen in login.png, the browser explicitly flags the connection as "Not Secure." The user enters the username admin and a password, successfully reaching the dashboard, which confirms the arrival over HTTP and displays sensitive financial data.

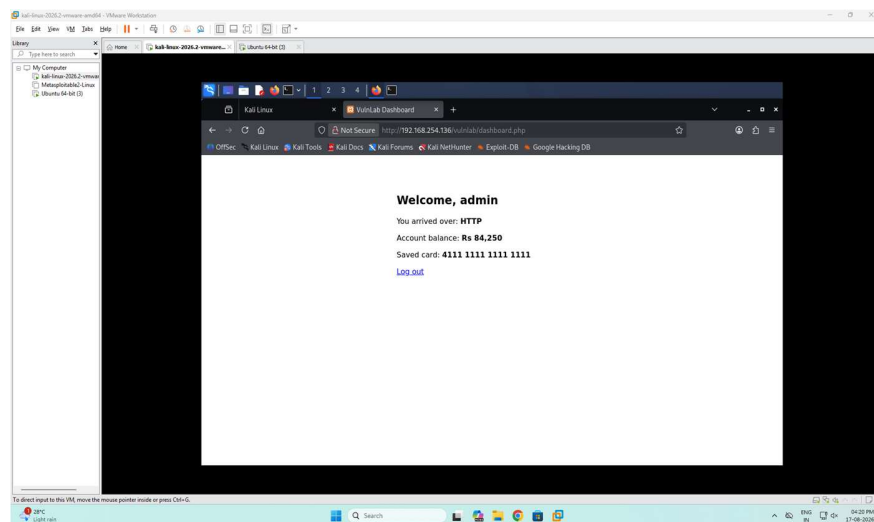


Fig 1: VulnLab dashboard accessed over insecure HTTP.

2. Packet Capture and Analysis

Using Wireshark to intercept the network traffic, we can filter for HTTP POST requests. This isolates the exact moment the user submitted their credentials to the server.

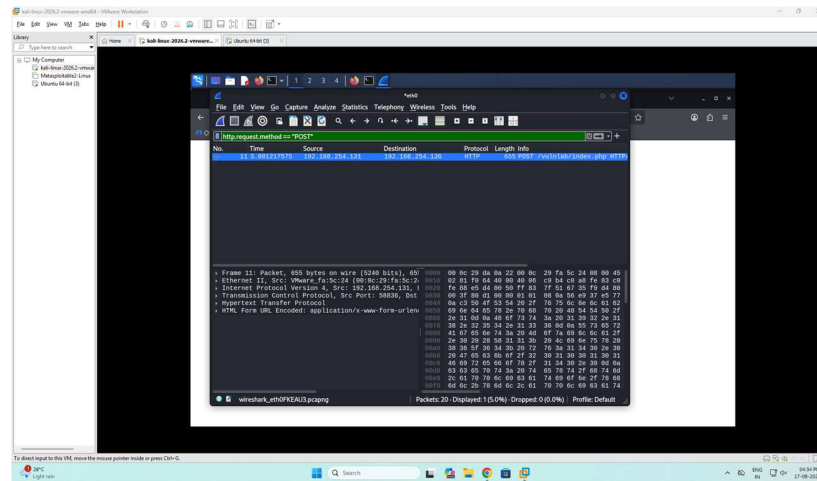


Fig 2: Wireshark capture filtered for HTTP POST requests.

3. The Vulnerability

By following the TCP stream of this HTTP POST request (follow http stream.jpg), the critical flaw of HTTP is exposed:

- **Plaintext Transmission:** The entire payload is sent in clear text.
- **Exposed Credentials:** The string `username=admin&password=SuperSecret123` is completely visible to anyone monitoring the network traffic. An attacker performing a Man-in-the-Middle (MitM) attack could easily harvest these credentials.

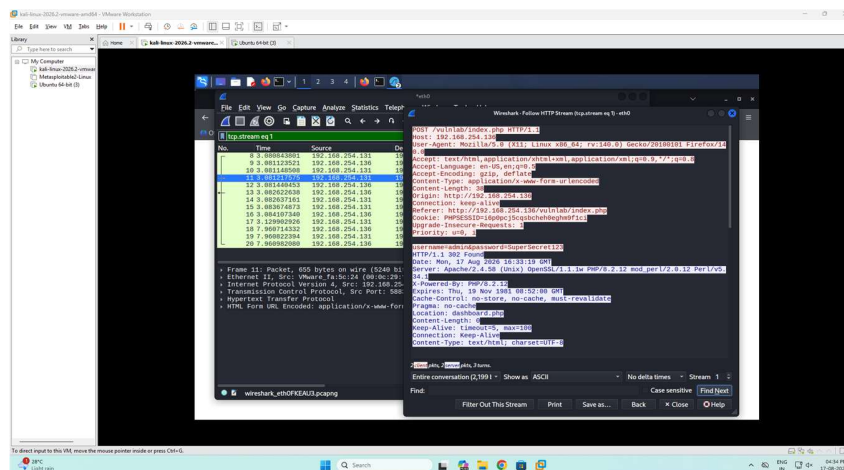


Fig 3: Wireshark capture filtered for HTTP POST requests.

Phase 2: HTTPS Traffic Analysis (Encrypted)

1. Establishing a Secure Connection

The user then attempts to access the same portal over HTTPS. Because the lab environment uses a self-signed certificate rather than one from a trusted Certificate Authority, the browser throws a "Potential Security Risk Ahead" warning.

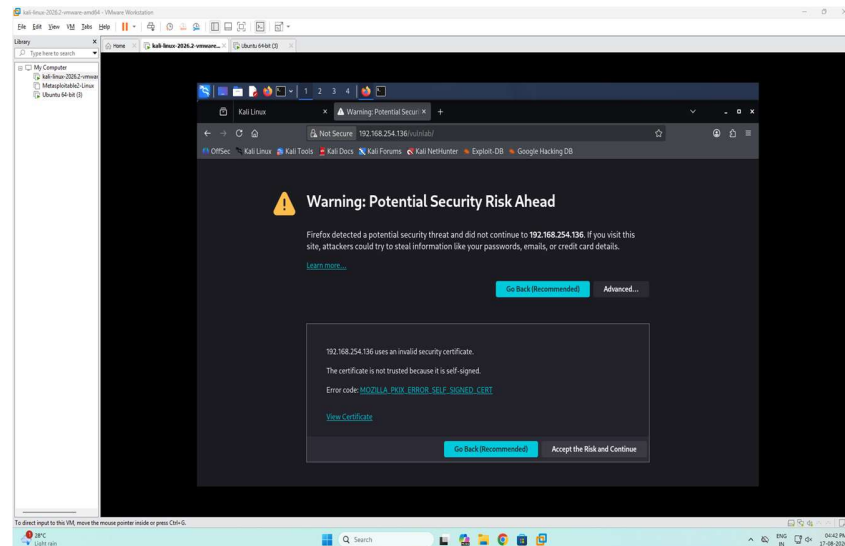


Fig 4: Firefox alerting to a self-signed certificate over HTTPS.

After accepting the risk, the user logs in and reaches the secure dashboard, confirming the arrival over HTTPS.

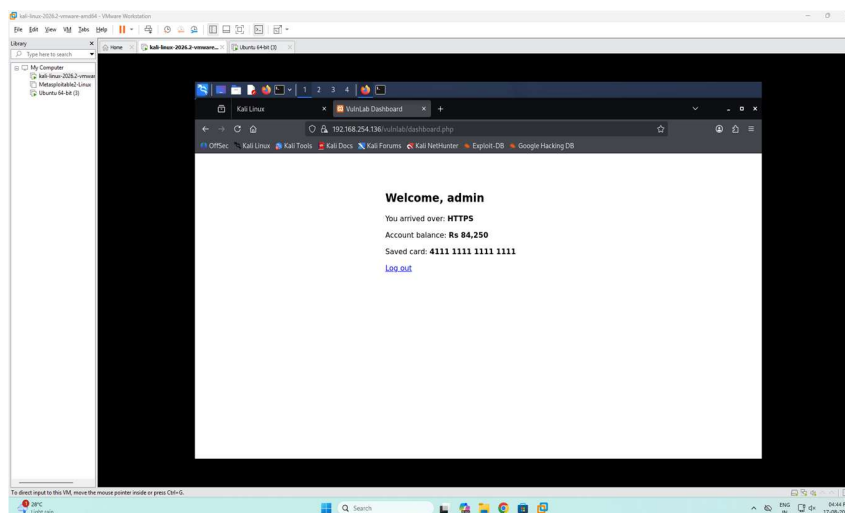


Fig 5: Secure login to the dashboard via HTTPS.

2. Packet Capture and Analysis

When repeating the Wireshark analysis on this new session, applying the same `http.request.method == "POST"` filter yields absolutely no results (second wireshark.png). This is because the HTTP traffic is now encapsulated within a secure tunnel.

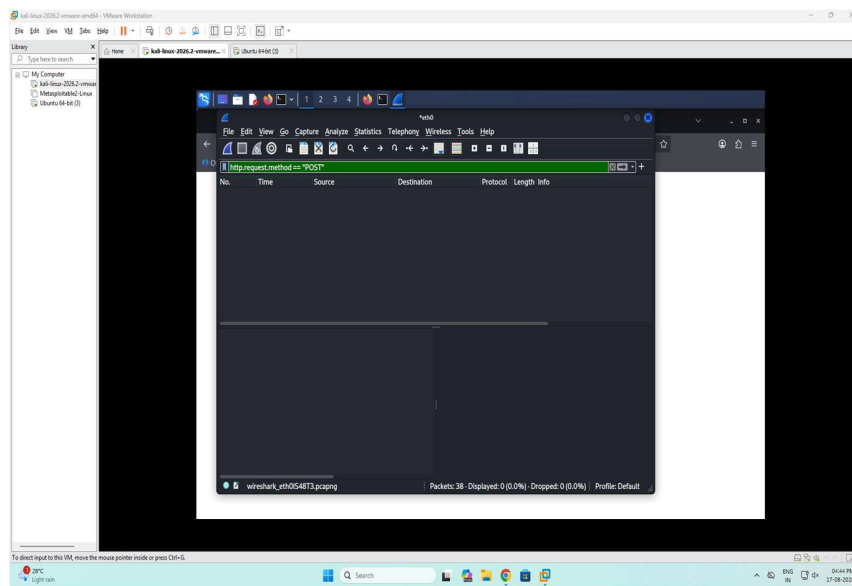


Fig 6: Wireshark showing no HTTP POST requests captured over HTTPS.

3. The TLS Handshake and Encryption

Instead of raw HTTP, Wireshark captures Transport Layer Security (TLS) traffic. This shows the client and server negotiating encryption parameters (Client Hello, Server Hello, Change Cipher Spec).

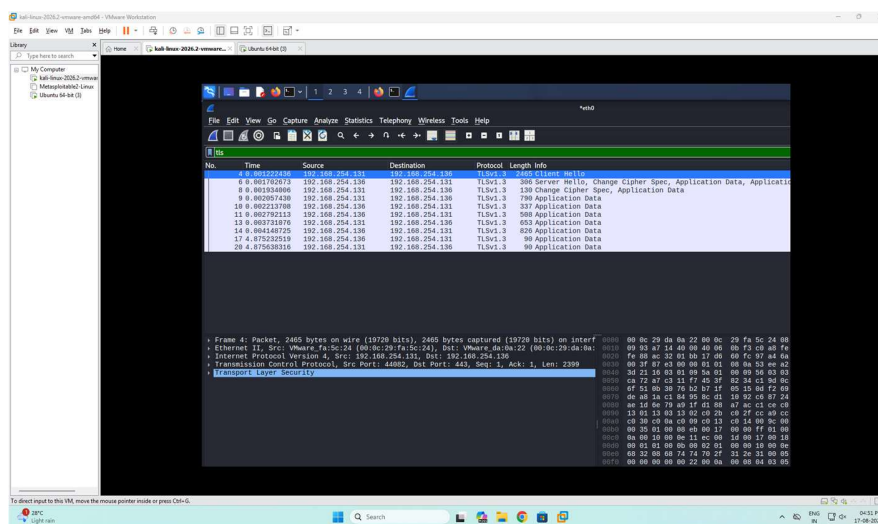


Fig 7: Secure communication established via the TLS protocol.

- **Encrypted Payload:** When inspecting the Application Data packets, the payload is completely unreadable. Wireshark labels it as Encrypted Application Data.

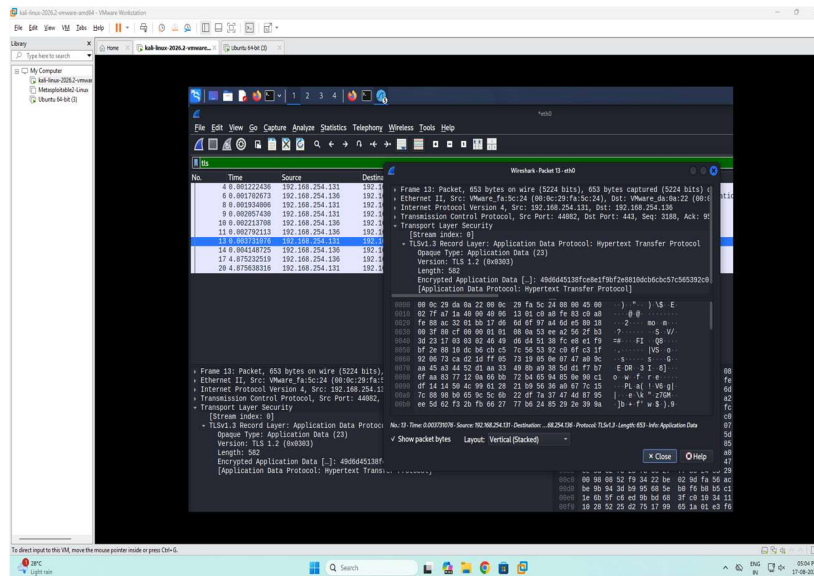


Fig 8: Wireshark displaying an unreadable encrypted payload.

- Data Protection:** To verify the credentials are truly hidden, a Wireshark search for the string frame contains "SuperSecret123" returns zero packets. The encryption successfully masks sensitive data.

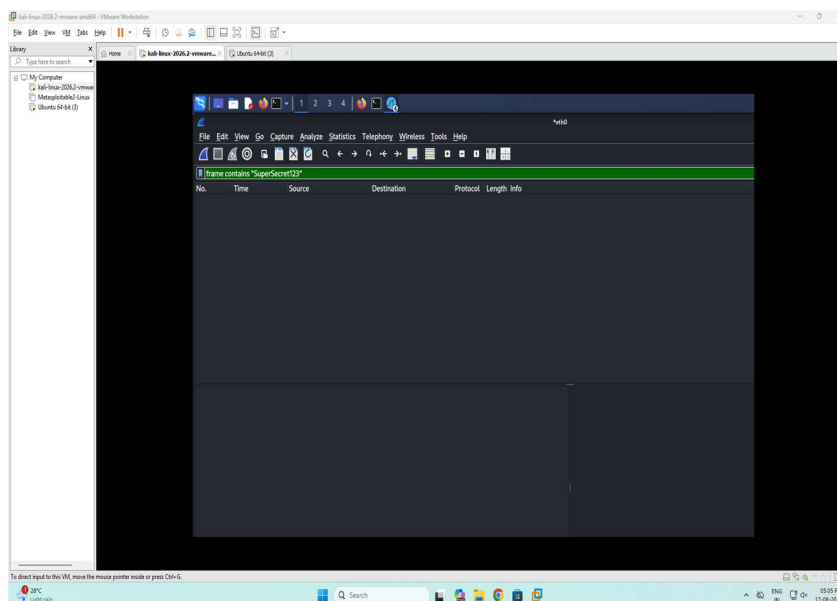


Fig 9: Wireshark search confirms the password is not exposed.